MIL-B-87031B(SH)
6 February 1987
SUPERSEDING
MIL-B-87031A(SH)
9 June 1978
(See 6.7)

MILITARY SPECIFICATION

BOOTS, DIVERS (MK-12)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a weighted rubber boot, with knurled calendered sole and built-in spike resistant steel midsole.

1.2 Classification. The boots shall be of one type and the following sizes as specified (see 6.2.1).

Sizes 8 - 10 - 12 - 14

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC FSC 4220

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MILITARY

MIL-S-5059 - Steel, Corrosion-Resistant (18-8), Plate, Sheet and Strip.
MIL-H-9890 - Hardware, Individual Load Carrying Equipment: and Hardware, Miscellaneous.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-284 - Visual Inspection Guide for Rubber Footwear.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20420.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z41 - American National Standard for Personnel Protection Protective Footwear.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Standard Test Methods for Rubber Properties in Tension. (DoD adopted)
D 573 - Standard Test Method for Rubber-Deterioration in an Air Oven. (DoD adopted)
D 624 - Standard Test Method for Rubber Property - Tear Resistance. (DoD adopted)
D 1630 - Standard Test Method for Rubber Property - Abrasion Resistance (NBS Abrader). (DoD adopted)
D 1870 - Standard Practice for Elevated Temperature Aging Using a Tubular Oven. (DoD adopted)
D 2240 - Standard Test Method for Rubber Property-Durometer Hardness. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials.

3.2.1 Rubber. The rubber compounds shall be made from a natural rubber, a synthetic rubber, or a mixture of the two to meet the requirements specified herein. Unless otherwise specified herein, the color shall be black (number 27038 in accordance with FED-STD-595). Where the color yellow is specified, it shall be color number 23655 in accordance with FED-STD-595.

3.2.1.1 Upper compounds.

3.2.1.1.1 Tensile strength. The tensile strength of the upper rubber compounds shall be not less than 1450 pounds per square inch (lb/in^2) (see 4.7.1).

3.2.1.1.2 Elongation. The ultimate elongation of the upper rubber compounds shall be not less than 350 percent (see 4.7.1).

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3.2.1.1.3 Accelerated aging. After aging, the upper rubber compounds shall have a tensile strength and elongation of not less than 75 percent of the minimum required (see 4.7.1).

3.2.1.2 Sole and heel compounds.

3.2.1.2.1 Abrasive index. The abrasive index of the rubber sole and heel compounds shall be not less than 35 to 45 before aging. After aging, the abrasive index shall be not less than 26, (see 4.7.1). The color shall be yellow (number 23655) for the outsole and black for the heel.

3.2.1.2.2 Hardness. The hardness of the rubber sole and heel compounds shall be 65 ± 5 shore-a-durometer before aging and shall be not more than 75 after aging (see 4.7.1).

3.2.2 Sponge rubber. When used, the sponge rubber shall be a self-blown type, calendered as applicable and activated or cured. The color is optional.

3.2.3 Fabrics. Fabrics to be coated for use in construction of the boots shall be dyed or undyed and shall contain not more than 0.003 percent copper and not more than 0.002 percent manganese (see 4.7.1). The color-fastness requirements shall not apply.

3.2.3.1 Cloth, osnaburg, cotton. The osnaburg fabrics shall be of two grades and shall be cotton, plain weave, unbleached or natural in color and shall conform to the minimum requirements of table I (see 4.7.1).

TABLE I. Osnaburg cotton cloth.

Grade	Weight per sq yd (ounces)	Yarns per inch		Nonfibrous [1] materials (percent)
		Warp	Filling	
1	2.56	34	24	12.0
2	3.65	34	24	12.0

[1] Starch and protein content, including chloroform soluble and water soluble material of the finished cloth.

3.2.3.2 Cloth, knitted cotton. The knitted cloth shall be of two grades, shall be cotton, natural or unbleached in color, and shall conform to the minimum requirements specified in table II (see 4.7.1).

TABLE II. Knitted cotton cloth.

Grade	Weight per sq yd (ounces)	Yarns per inch	
		Wales	Courses
1	5.2	16	19
2	4.3	24	32

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3.2.3.3 Cloth, knitted (nylon and cotton). The knitted cloth used for the leg lining shall be of a nylon face and cotton backing. There shall be a minimum of 54 percent nylon and 46 percent cotton by weight of the finished fabric. The fabric may be dyed any color or may be undyed and shall conform to the requirements of table III (see 4.7.1).

TABLE III. Knitted cloth (nylon and cotton).

Weight per sq yd (ounces)	Yarns per inch	
	Wales	Courses
5.0	22	25

3.2.3.4 Felt, nylon. The felt used for the toe plug shall be nylon, a minimum of 3/8 inch thick, and shall be of any suitable color. The colorfastness requirements shall not apply.

3.2.3.5 Duck, cotton. The cotton duck used to cover the steel midsole shall be natural or unbleached, 7.0 ounces per square yard with a minimum of 28 warp ends per inch and 20 picks per inch (see 4.7.1).

3.3 Cord, reinforcement. The cord used for reinforcing the edges of the buckle, loop, and adjustment straps shall be cotton, natural in color and shall be size 12/4/2 (see figure 1). As an alternate, any corded yarn of any size and material may be used.

3.4 Cord, release. The release cord used as a pull tab for the buckle shall be nylon, polyester or cotton, hollow or solid braided, approximately 1/8 inch in diameter and a minimum breaking strength of 75 pounds. The cord shall be natural or colored and colorfastness requirements shall not apply (see figure 1).

3.5 Metal components.

3.5.1 Steel box toe. The steel box toe shall conform to the toe of the last and shall be fabricated from cold-rolled carbon steel. After heat treatment, the steel box toe shall conform to the requirements specified in table IV. The steel box toe shall be thoroughly cleaned and given a zinc compound coating or other suitable coating.

TABLE IV. Physical requirements, steel box toe.

Hardness (Rockwell C)	Carbon content (percent)	Thickness (inch)
42-52	0.50 to 0.82	0.050 to 0.058

3.5.1.1 Impact resistance. The steel box toe of the finished boot shall have a minimum inside clearance of 1/2 inch and shall show no evidence of fracture (see 4.5.3.2).

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3.5.2 Steel sole. Steel for the insole plate shall be corrosion resistant, composition 301 or 302, condition hard, finish 2D or 2B, as specified in MIL-S-5059.

3.5.2.1 Puncture resistance. The steel insole shall have a minimum puncture resistance of 200 pounds (see 4.7.2). Any single determination below 200 pounds shall cause rejection of the lot.

3.5.3 Buckles. The quick release buckle shall conform to type XIV of MIL-H-9890 (see 6.4 and figure 1). In addition, replacement buckles shall be furnished with marks as shown on figure 2.

3.5.3.1 Replacement of buckles. When defective buckles are to be replaced on the boot, ends of replacing buckles shall be cut on scribe marks and filed smooth (see figure 2). Ends of buckles shall be inserted on strap loops. Buckles for boots being manufactured do not require the cutout as shown on figure 2.

3.5.4 Loops. Loops used for the boot closure shall be steel or brass wire, with a chrome or rustproof finish, 1/8 inch in diameter. Inside dimensions of the loop opening shall be 1 by 1/2 inch (see 6.4 and figure 1).

3.5.5 Lead. The lead used for the heel weight shall be a pure commercial caulking lead. The weight of lead added to each boot shall be a minimum of 2 pounds and a maximum of 2-3/4 pounds. When specified in the contract or order, a certificate of compliance shall be prepared for this requirement (see 6.2.2).

3.5.6 Rivets and washers. Rivets and washers used to attach the lead heel weight to the steel midsole shall be brass.

3.6 Cement. Cement used for sealing the cut edges of the front opening shall be a self-curing type.

3.7 Lasts and patterns. The lasts and patterns shall be used.

3.8 Construction. Boots shall be constructed in accordance with the patterns and lasts, using the component parts specified herein. Overlapped parts shall be firmly rolled or pressed before vulcanization.

3.8.1 Leg lining. The leg lining shall consist of the material specified in 3.2.3.3. It shall be calender coated on the cotton back side with upper compound not less than 0.009 inch thick.

3.8.2 Toe lining. The toe lining shall consist of grade 1 material specified in 3.2.3.2. It shall be calender coated on one side with upper compound not less than 0.015 inch thick.

3.8.3 Leg form. The leg form shall consist of the grade 1 material specified in 3.2.3.1. It shall be friction coated on both sides and calender coated on one side with upper compound not less than 0.008 inch thick.

3.8.4 Front strip. The front strip shall be grade 1 material specified in 3.2.3.1, friction coated on both sides. The strip shall be a minimum of 3/4 inch wide.

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3.8.5 Midsole covers. The steel midsole covers (top and bottom) shall be of the material specified in 3.2.3.5, friction-coated on both sides and calender coated on one side with upper compound not less than 0.007 inch thick.

3.8.6 Vamp reinforcement. The vamp reinforcement shall be grade 2 material specified in 3.2.3.2, spreader coated and calender coated on both sides with upper compounds with a total thickness of 0.015 inch.

3.8.7 Insole. The insole shall consist of grade 2 material as specified in 3.2.3.1, coated on one side with cork compound not less than 0.080 inch thick, then calender coated not less than 0.006 inch thick, and calender coated with rag compound (stiffening compound) not less than 0.040 inch thick.

3.8.8 Backstay strip. The backstay strip shall be grade 1 material as specified in 3.2.3.1 and friction coated on both sides. The strip shall be a minimum of 3/4 inch wide.

3.8.9 Friction toe reinforcement. The toe reinforcement shall be grade 1 material specified in 3.2.3.1 and friction coated on both sides.

3.8.10 Friction heel. The friction heel shall be of grade 1 material as specified in 3.2.3.1 and friction coated on both sides.

3.8.11 Friction holding tape. The friction holding tape shall be of grade 1 material as specified in 3.2.3.1, friction coated on both sides and have a minimum width of 1-1/2 inches.

3.8.12 Midsole tie in strip. The midsole tie in strip shall be of grade 1 material as specified in 3.2.3.1 and friction coated on both sides. The strip shall be a minimum of 3/4 inch wide.

3.8.13 Vamp form. The vamp form shall be grade 2 material as specified in 3.2.3.2, spreader coated and then calender coated with upper compounds on both sides with a total thickness of 0.015 inch.

3.8.14 Binder sole. The binder sole shall be grade 2 material as specified in 3.2.3.1, friction coated on both sides, then calender coated on one side with rag compound not less than 0.045 inch thick.

3.8.15 Buckle straps. Straps to secure the buckles and loops to the boot shall be made of grade 1 material as specified in 3.2.3.1, friction coated on both sides and calender coated with upper compound on one side with a thickness of 0.011 inch. Each strap shall be cut 2-1/2 ± 1/4 inches long by 2 ± 1/4 inches wide. The straps shall be folded on the length direction to finish 1 ± 1/16 inch wide. Each edge of the folded strap shall have a reinforcement cord specified in 3.4.

3.8.16 Adjustment strap (top). The top adjustment strap shall be made of grade 1 material as specified in 3.2.3.1, friction coated on both sides and calender coated on one side with upper compound not less than 0.011 inch thick. The strap shall be cut 17 ± 1/4 inches long by 2 ± 1/4 inches wide. The strap shall be folded on the length direction to finish 1 ± 1/16 inch wide. Each edge of the folded strap shall have a reinforcement cord as specified in 3.3. The free end

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of the strap shall have a tapered cut end to facilitate threading through the buckle (see figure 1). Each pair of boots shall be furnished with two separate, additional top straps.

3.8.17 Adjustment strap (bottom). The bottom adjustment strap shall be constructed as specified in 3.8.16, except that the strap shall be cut 14 ± 1/4 inches long (see figure 1). Each pair of boots shall be furnished with two separate, additional bottom straps.

3.8.18 Back loop strip. The back loop strip shall be constructed as specified in 3.8.16, except that the strap shall be cut 3 ± 1/4 inches long (see figure 1).

3.8.19 Steel midsole. The steel midsole shall be as specified in 3.5.2. The steel sole shall be formed to fit the contour of the bottom of the last. The midsole shall be coated with a protective coating to prevent corrosion and cement coated for further fabrication in the boot.

3.8.20 Felt toe plug. The felt toe plug shall be of the material as specified in 3.2.3.4, calender-coated on both sides not less than 0.010 inch thick. The felt shall fill the area of the boot bottom forward of the ball line and edge of lead heel weight.

3.8.21 Heel weight. Material for the heel weight of each boot shall be as specified in 3.5.5. The lead weight shall approximately fill the heel area and shank area to just rear of the ball line of the boot.

3.8.22 Filler sole. The filler sole shall be a stiffening compound (plain rag), 0.070 ± 0.005 inch in thickness.

3.8.23 Buckle strap, reinforcement. The gum rubber reinforcement straps to secure the buckles and loops to the boot shall be of upper compound as specified in 3.2.1.1, not less than 0.0375 inch thick. The strap shall be 1 inch wide by 2 inches long (see figure 1).

3.8.24 Leg cover. The gum leg cover shall be of upper compounds as specified in 3.2.1.1, not less than 0.0375 inch in thickness.

3.8.25 Backstrip. The gum backstrip shall be of upper compound as specified in 3.2.1.1, not less than 0.0375 inch in thickness.

3.8.26 Toe reinforcement (sponge). The sponge toe reinforcement shall be as specified in 3.2.2, not less than 0.050 inch in thickness before vulcanization.

3.8.27 Heel reinforcement. The gum heel reinforcement shall be made of upper compounds specified in 3.2.1.1, not less than 0.0375 inch in thickness.

3.8.28 Piping. The half-round piping, used to cushion the lead heel weight along its outside edges, shall be of upper compound as specified in 3.2.1.1, not less than 0.0100 inch in thickness.

3.8.29 Gum heel. The gum heel shall be of upper compound as specified in 3.2.1.1, not less than 0.0375 inch in thickness.

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3.8.30 Gum heart. The gum heart used to reinforce the strap for the top loop (back center of boot) shall be of upper compound as specified in 3.2.1.1, not less than 0.0375 inch in thickness.

3.8.31 Gum vamp. The gum vamp shall be of upper compound as specified in 3.2.1.1, not less than 0.060 inch in thickness.

3.8.32 Bumper toe cap. The bumper toe cap shall be of upper compound as specified in 3.2.1.1, not less than 0.045 inch in thickness. Color shall be yellow to match the outsole; foxing and embossed finish is optional.

3.8.33 Foxing. Foxing shall be of upper compound as specified in 3.2.1.1, not less than 0.065 inch thick. Color shall be yellow to match the outsole; bumper toe cap and embossed design is optional.

3.8.34 Binding. The binding shall be of upper compounds as specified in 3.2.1.1, not less than 0.090 inch thick. Color and inscription are optional.

3.8.35 Outsole. The outsole shall be made of the compound specified in 3.2.1.2. Minimum thickness shall be 0.375 inch at the ball area and 0.150 inch at the heel seat area before cementing of the heel. Color, knurled tread design and inscription are optional. The outsole shall have an instep rise of approximately 1-1/4 inches and an outside rise of approximately 1-1/2 inches.

3.8.36 Heel. The heel shall be molded of the compound specified in 3.2.1.2. Height of the heel shall be 1 inch at back center (including tread design). The color shall be optional. The heel shall be cemented to the heel area with a adhesive prior to vulcanization.

3.9 Measurements. The inside height of the finished boot in the heel seat area shall be a minimum of 8-1/4 inches. After vulcanization, the front of the boot shall have a "V" cut approximately 2 inches at the top edge tapering to a point at 7 to 7-1/4 inches from the top edge (see figure 3).

3.10 Marking.

3.10.1 Size marking. The following size information shall be applied to the boots. Location and sizing nomenclature are detailed below for the four size numbers required.

Location	Nomenclature
(a) Outside of each boot in shank area of each boot's outsole.	Number size not less than 0.5 inch

Industry size	Size markings on boot
8	8/1
10	10/2
12	12/3
14	14/4

Size marking on boot outsole shall be embossed, branded, or incised into the outsole in the shank area of each boot.

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(b) The outside quarter and inside quarter of each boot shall have size numbers vulcanized permanently into the leg cover material. The size numbers shall be yellow rubberized material die cut for a full 1.5 inch high Arabic numeral. The yellow color shall be a commercial quality high visibility safety yellow or equivalent, in accordance with FED-STD-595. The numbers are not to be inked on the boot and sealed over. As noted above, the yellow rubberized material numbers shall be vulcanized on the boot side quarters as a patch would be vulcanized on a tire. The number shall be oriented vertically for reading with the boot situated upright as on a shelf or table.

The numbering shall be as follows:

Industry size	Boot size number
8	1
10	2
12	3
14	4

3.10.2 Labels. The label markings shall contain the following information, legibly stamped or printed with indelible ink of a contrasting color on the inside of the boot approximately 1 inch below the top edge. The size marking shall be a minimum of 1/2 inch in height. Other markings shall be a minimum of 3/16 inch.

"(SIZE)
BOOT, DIVERS (MK-12)
NAME OF CONTRACTOR:
CONTRACT NUMBER:"

3.11 Vulcanization. Boots shall be vulcanized under pressure to ensure that parts and seams are securely adhered to each other to form a single unit. Seams on outside of boot shall be rolled prior to vulcanization.

3.12 Workmanship. The finished boots shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL) (see 4.5.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. When specified in the contract or order a certificate of compliance shall be prepared for the material requirements specified in 3.5.5 (see 6.2.2).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article submitted in accordance with 3.1 shall be inspected as specified in 4.5.3.1 for compliance with design, construction, workmanship and dimensional requirements and tested as specified in 4.5.3.2 and 4.7. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2.2).

4.4 Quality conformance inspection. Unless otherwise specified herein, sampling and inspection shall be performed in accordance with MIL-STD-105. For purpose of sampling, an inspection lot for quality conformance inspection shall consist of all material of the same type offered for delivery at one time. Lot and sample size shall be as specified in table V, and tested in accordance with 4.7.1.

TABLE V. Sample size.

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.1 The lot shall be unacceptable if one or more sample units or the composite sample fail to meet any requirement specified. A lot shall consist of the amount of rubber compound that is produced or compounded at any one time. The composite test sample of rubber shall be vulcanized in the same manner employed in the manufacture of boots. For each type of rubber compound, the sample unit shall be a composite of the rubber prepared in the form as follows:

(a) Upper compounded rubber-two pieces 9 by 9 by 0.070 ± 0.010 inch.
(b) Sole and heel compounded rubber-two pieces 6 by 6 by 0.070 ± 0.010 inch and one piece 6 by 6 by 0.250 ± 0.025 inch.

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4.5 Examinations.

4.5.1 Inspection levels and acceptable quality levels. The inspection levels and AQL's in accordance with MIL-STD-105, expressed in defects per 100 units, shall be as follows:

	Inspection level	AQL's Major	AQL's Total
For defects applicable to 4.5.2	II	2.5	6.5
For defects applicable to 4.5.3	S-3		4.0

4.5.2 Visual examination. The finished boots shall be examined for the defects specified in table VI and the applicable defects as specified in MIL-STD-284. The sample unit shall be one completely fabricated boot and the selection shall be in pairs. For pairing examination, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect.

TABLE VI. Visual defects.

Item number	Defect
Major	
101	Color of component parts not as specified.
102	Any heel not as specified.
103	Any component missing or not as specified.
104	Height of heel less than specified.
105	Buckle missing.
Minor	
201	Overlapped parts or seams on the outside not rolled or pressed.
202	Dimensions of any component not as specified.
203	Marking incorrect.
204	Additional straps as specified, missing.

4.5.3 Dimensional examination. The boots shall be examined for defects in dimensions. Dimensions not within the established tolerance shall be classified as defects. The sample unit shall be one boot.

4.5.3.1 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.5.2 and 4.5.3. The inspection level and AQL shall be as specified in 4.5.1.

4.5.3.2 Testing of the end item. The finished boots shall be tested for performance characteristics specified in table VII. The sample unit for the impact test shall be one boot and the selection shall be by pairs. For the impact test, the inspection level shall be S-1 and the AQL shall be 4.0 defects per one hundred units.

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TABLE VII. End item meeting.

Characteristic	Specification reference		Requirement applicable to		Number of determinations per individual unit of product	Results reported as
	Requirement paragraph	Test method	Individual unit of product	Lot average		
Impact	3.6.1.1	4.7.2 or ANSI Z41	X		1	Pass or fail

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.6.1 Sampling. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed in table VIII shall be examined on shipping containers fully prepared for delivery.

4.6.2 Lot size. The lot size shall be the number of containers in the end item inspection lot. The inspection level shall be S-2 in accordance with MIL-STD-105 and the AQL shall be 2.5 defects per 100 units.

TABLE VIII. Packaging defects.

Examine	Defects
Marking (exterior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application. Size on item marking not in conformance with size shown on exterior container. [1]
Materials	Any component missing, any component damaged. Any component not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling. Bulging or distortion of containers.
Content	Number of pairs of boots per container is more or less than specified.

[1]For this defect, one pair from each container shall be examined.

4.7 Tests.

4.7.1 Testing of components. In addition to the quality assurance provisions of the referenced specifications, components specified in table IX shall be tested for the characteristics shown therein.

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TABLE IX. Testing of components.

Component and unit of product	Characteristic	Specification reference		Requirement applicable to		Number of determinations per sample unit	Results reported as	Sample unit
		Requirement paragraph	Test method	Sample unit	Lot average			
Upper compound	Tensile strength: Before aging After aging	3.2.1.1.1 and 3.2.1.1.3	D 412[2] [4] D 412, D 573 and D 1870[2] [3] [4]	---	---	---	---	---
	Elongation: Before aging After aging	3.2.1.1.2 and 3.2.1.1.3	D 624[2] D 624, D 573 and D 1870[2] [3]	---	---	---	---	---
Sole and heel compound	Abrasive index Before aging After aging	3.2.1.2.1	D 1630[2] D 1630[2] D 573 and D 1870[2] [4]	---	---	---	---	---
	Hardness: Before aging After aging	3.2.1.2.2	D 2240[2] D 2240, D 573 and D 1870[2] [3]	---	---	---	---	---
Osnaburg (cotton)	Material identification	3.2.3.1	1200[1]	---	---	---	---	1 yard full width
	Wgt/oz/sq/yd	3.2.3.1	5041[1]					
	Yards per inch	3.2.3.1	5050[1]					

See footnotes at end of table.

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TABLE IX. Testing of components. - Continued

Component and unit of product	Characteristic	Specification reference		Requirement applicable to		Number of determinations per sample unit	Results reported as	Sample unit
		Requirement paragraph	Test method	Sample unit	Lot average			
Osnaburg (cotton)	Warp Filling Weave Non-fibrous Copper and manganese content	 3.2.3.1 3.2.3.1 3.2.3	 1200[1] 2611[1] 2053[1]	 X	 ---	 2	 nearest 0.001 percent	
Duck (cotton)	Material identification Wgt/oz/sq yd Weave Copper and manganese content	3.2.3.5 3.2.3.5 3.2.3.2 3.2.3	1200[1] 5041[1] Visual 2053[1]	 X	 ---	 2	 nearest 0.001 percent	1 yard full width
Knit (cotton)	Material identification Wgt/oz/sq yd Copper and manganese content	3.2.3.2 3.2.3.2 3.2.3	2053[1] 5041[1] 2053[1]	 X	 ---	 2	 nearest 0.001 percent	1 yard full width
Knit (nylon and cotton)	Material identification Face (percent) Back (percent) Wgt/oz/sq yd Texture: Wales per inch Courses per inch	3.2.3.3 3.2.3.3 3.2.3.3	 5041[1] 5050[1]	---	---	---	---	1 yard full width
Steel sole	Puncture resistance	3.5.2.1	4.7.2	X	---	2	pounds	3 steel toes

See footnotes at top of next page.

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[1] Refers to test method 1200, 2053, 2611, 5041 or 5050 of FED-STD-191.
[2] Refers to ASTM D 1630, D 412, D 573, D 624, D 1870 and D 2240.
[3] Sample shall be aged for 16 hours at 212 ± 2 degrees Fahrenheit.
[4] Tensile strength and elongation may be performed simultaneously.

4.7.2 Puncture resistance test.

4.7.2.1 Apparatus. The puncture resistance test shall be performed on a testing machine having a movable platform adjusted to travel at 1/4 inch per minute. Two blocks of hardwood, metal, or plastic shall be prepared as follows: the blocks shall be of such size and thickness as to ensure a suitably rigid test ensemble and allow for at least 1 inch of the pointed end of an 8D nail to be exposed for the penetration. One block shall have a hole drilled to hold an 8D common nail firmly at an angle of 90 degrees. The second block shall have a maximum 1/2 inch diameter hole drilled through it so that the hole will allow free passage of the nail after it penetrates the insole during the test.

4.7.2.2 Procedure. The test ensemble consisting of the sample unit, the two prepared blocks, a piece of leather outsole 10 to 11 irons thick, and a new 8D nail, shall be placed as follows: the 8D nail in the hole, the sample of outsole stock superimposed above the nail, the area of the sole plate to be tested placed on the outsole, and the second block with hole so placed as to allow for free passage of the nail after it passes through the outsole stock and sole plate in that order. The machine shall be started and the pressure, in pounds required for the nail to completely penetrate the outsole and sole plate, recorded to the nearest 5 pounds. Two determinations shall be made on each sole plate and the results averaged. A new nail shall be used for each determination.

4.7.3 Impact. The impact test shall be performed in accordance with the following procedure: A falling weight of steel or other material, weighing not less than 40-1/2 pounds nor more than 50-1/2 pounds, 1 ± 0.020 inch in diameter and with a striking face made hemispherical in shape by rounding on a 1 inch radius, equipped to drop freely in a tube or other suitable guide, shall be used. The weight shall be dropped from a distance of 18 ± 1/4 inches above the top of the toe-box. As an alternate, a falling weight of 25 ± 14 pounds shall be dropped from a distance of 36 ± 1/2 inch. The equipment shall be assembled so that the center of striking face of the falling weight will strike the boot at midwidth 1/2 inch in front of the back edge of the toe-box. The boot (or toe-section) shall be mounted on an anvil (or rigidly mounted bedplate) of steel or other suitable material weighing not less than 500 pounds, equipped with guides (or clamps) to hold the boot specimen in place. In performing the test, the instantaneous clearance may be determined by placing lumps of a plastic material such as wax or modeling clay inside the boot directly below the point of impact. The plastic material should be compressed with the fingers so that it contacts both upper and lower inside surfaces of the boot before the test is made. When in place, the greatest horizontal dimension of the lumps should not exceed 1 inch. Other methods for determining the instantaneous clearance may be used.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2.1).

5.1.1 Level A. Each pair of mated boots with extra top and bottom straps shall be placed in an open end clear polyethylene bag 0.003 inch thick. Dimensions of the bag shall be the option of the contractor.

5.1.2 Commercial. Finished boots shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level B, C or commercial as specified (see 6.2.1).

5.2.1 Level B. Six pairs of boots of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type CF, class weather-resistant, variety SW, grade V3c of PPP-B-636. Each box shall be fitted with a liner conforming to type CF, class weather-resistant, variety SW, grade V3c of PPP-B-636. Inside dimensions of the shipping container shall be approximately 33 inches long, 16 inches wide by 13 inches deep. Approximate dimensions are furnished as a guide only. Each container shall be closed and reinforced in accordance with the appendix of the container specification. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2 Level C. Six pairs of boots, of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type CF, class domestic, variety SW, grade 200 or 275 of PPP-B-636. Each container shall be fitted with a liner conforming to type CF, class domestic, variety SW, grade 200 or 275 of PPP-B-636. Inside dimensions of the shipping container shall be approximately 33 inches long, 16 inches wide by 13 inches deep. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with the appendix of PPP-B-636. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed in the same container.

5.2.3 Commercial. Boots preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required (see 6.2.1), interior packs and exterior shipping containers for levels A, B and C shall be marked in accordance with MIL-STD-129. Commercial interior packs and exterior shipping containers shall be marked in accordance with ASTM D 3951.

5.3.1 Labels, mixed sizes. Each shipping container packed with mixed sizes only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words

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"MIXED SIZES" plainly stamped or printed thereon, and under these words shall be legibly stamped or printed the correct quantity of pairs and sizes contained therein.

6. NOTES

6.1 Intended use. The MK-12 divers boot is to be used in conjunction with the MK-12 diving system.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sizes required (see 1.2).
(c) Whether first article sample is required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.5.5 and 4.1.2	Certificate of compliance	DI-E-2121	----
4.3	First article inspection report	DI-T-4902	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 The quick-release buckle specified in 3.5.3 may be obtained from Prentice Corp., Kensington, CN 06037, Part No. 3832. The loops may be obtained from:

Wilcox-Crittenden (North & Judd)
P.O. Box 111
Middletown, CN 06457
Part No. 595 - 1 inch wide by 1/2 inch high - 0.135 diameter steel wire

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Boot, diver's
Buckle
Compounds, rubber
Cloth, OSNABURG
Cloth, knitted
Strap, loop
Toe, steel box

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
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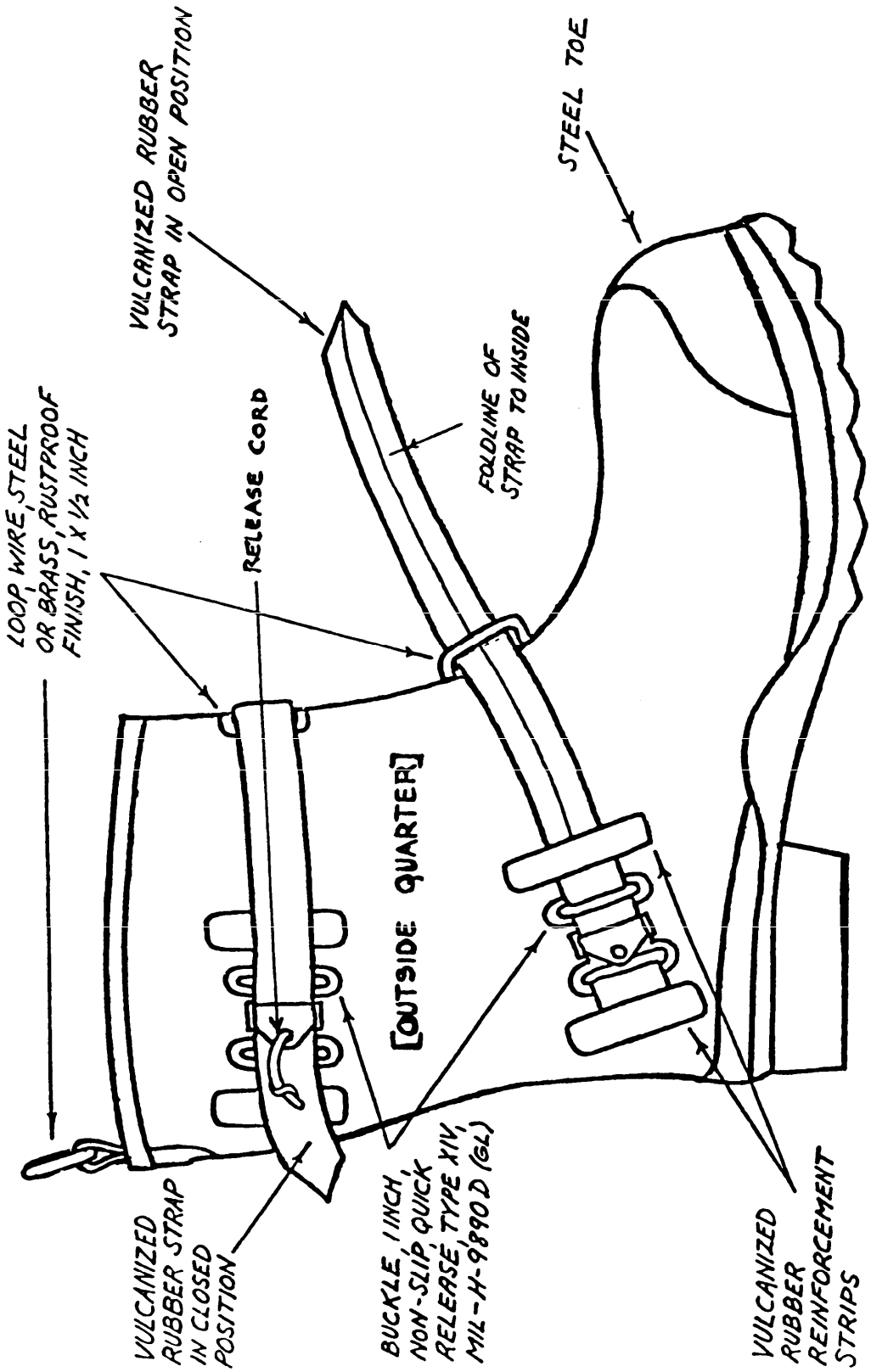
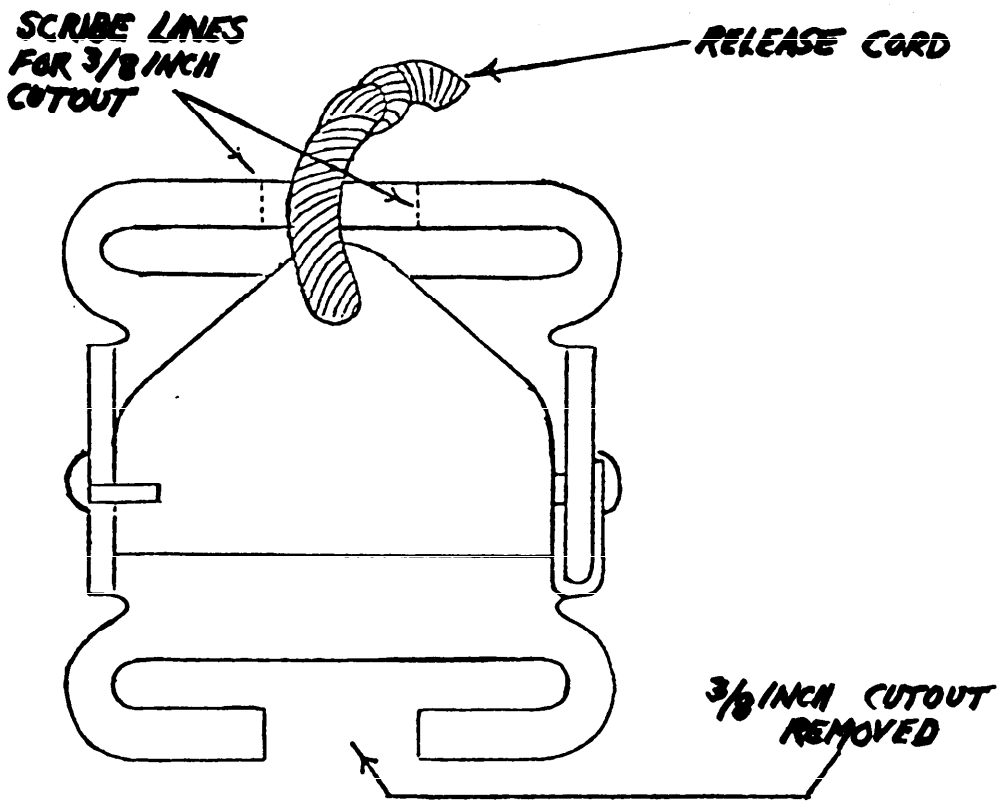


FIGURE 1. Boot, diver's (MK-12), side view.

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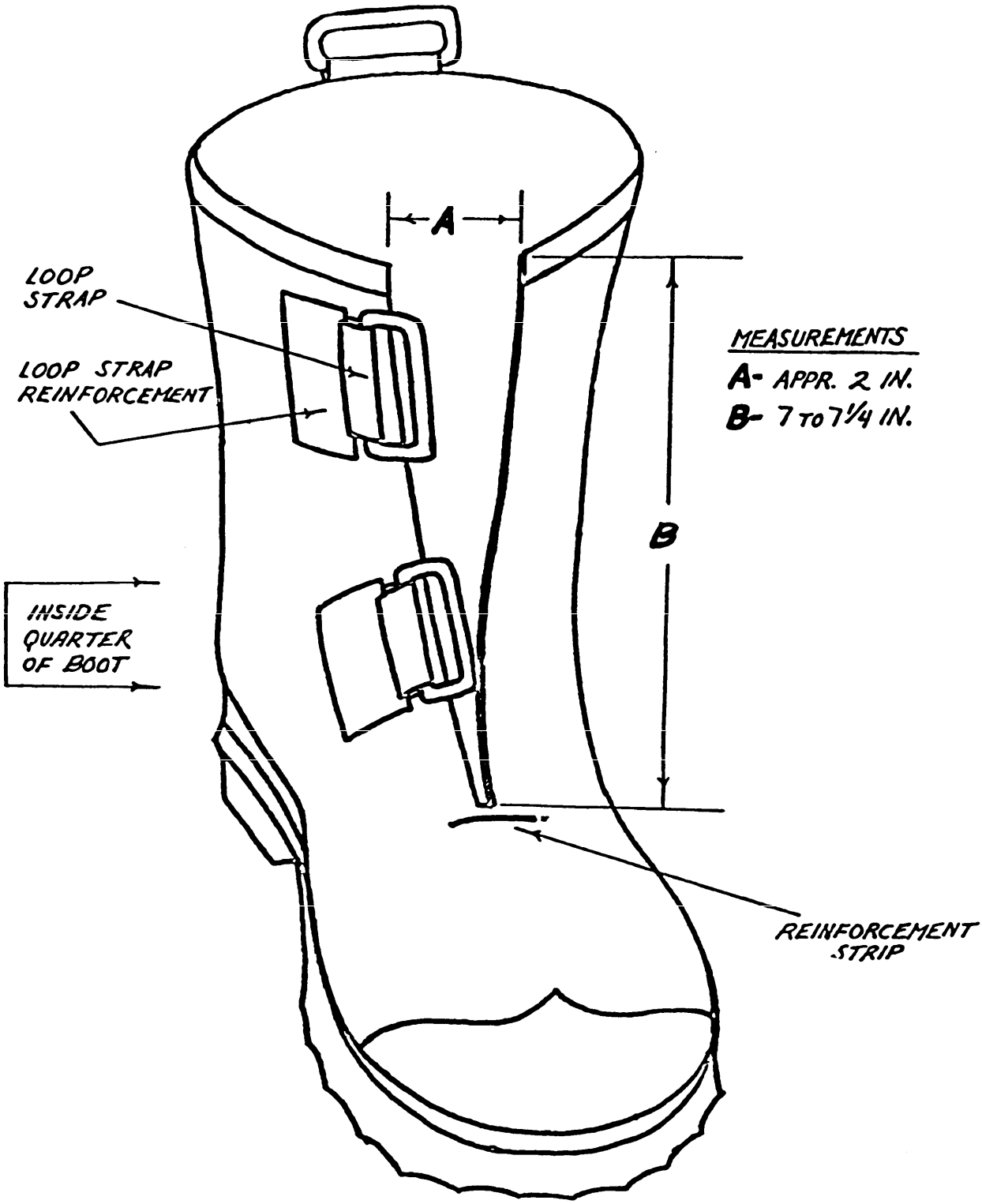
NOTE:

1. When attached, the pointed end of the tongue release shall face towards back of boot.

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FIGURE 2. Replacement buckle for boot, diver's (MK-12).

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NOTES:

1. Adjustment straps and buckles are not shown on this illustration.
2. Refer to guide sample for proper placement of loops and straps.

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FIGURE 3. Boot, diver's (MK-12), front view.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-87031B(SH)	BOOTS, DIVERS (MK-12)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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